

EXPERIMENT 8

Demonstration of predictive models such as Decision Tree, Neural network and KNearest Neighbor using various domain datasets.

AIM

To implement and evaluate a Neural Network (MLPClassifier) on the Dry Bean Dataset and the Phishing Websites Dataset, and assess model performance using accuracy, classification report, confusion matrix, and ROC curves.

PROCEDURE

1. Install ucimlrepo library using pip.
2. Import ucimlrepo, pandas, matplotlib, sklearn modules.
3. Fetch the Dry Bean dataset (id=602) using fetch_ucirepo().
4. Extract features (X) and targets (y).
5. Encode target labels using LabelEncoder.
6. Split dataset into 80% train and 20% test sets.
7. Normalize features using StandardScaler.
8. Define MLPClassifier with layers (150, 100, 50), relu activation, adam solver.
9. Train the model using model.fit().
10. Predict on test set and compute accuracy.
11. Print classification report with target class names.
12. Plot confusion matrix using ConfusionMatrixDisplay.
13. Plot training loss curve.
14. Compute and plot ROC curves with AUC for each class.
15. Repeat steps 3–14 for the Phishing Websites dataset (id=327).

CODE

```
pip install ucimlrepo

from ucimlrepo import fetch_ucirepo
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import classification_report, accuracy_score, ConfusionMatrixDisplay

# Dataset 1: Dry Bean
dry_bean = fetch_ucirepo(id=602)
X = dry_bean.data.features
y = dry_bean.data.targets

y = y.ravel()
le = LabelEncoder()
y = le.fit_transform(y)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = MLPClassifier(
    hidden_layer_sizes=(150, 100, 50),
    activation='relu',
```

```

        solver='adam',
max_iter=500,
learning_rate_init=0.001,
random_state=42
)
model.fit(X_train, y_train)

y_pred = model.predict(X_test) print('Accuracy:',
accuracy_score(y_test, y_pred))

target_names_list = ['Barbunya', 'Bombay', 'Cali', 'Dermosan', 'Horoz', 'Seker', 'Sira']
print(classification_report(y_test, y_pred, target_names=target_names_list))

ConfusionMatrixDisplay.from_estimator(model, X_test, y_test)
plt.title('Confusion Matrix - Dry Bean Dataset') plt.show()

plt.plot(model.loss_curve_)
plt.title('Training Loss Curve')
plt.xlabel('Iterations')
plt.ylabel('Loss')
plt.show()

```

OUTPUT

Dataset 1 - Dry Bean:

Accuracy: 0.9243481454278369

Classification Report:

	precision	recall	f1-score	support
Barbunya	0.94	0.90	0.92	261
Bombay	1.00	1.00	1.00	117
Cali	0.91	0.96	0.93	317
Dermosan	0.90	0.92	0.91	671
Horoz	0.97	0.95	0.96	408
Seker	0.97	0.94	0.95	413
Sira	0.87	0.88	0.87	536
accuracy			0.92	2723

Dataset 2 - Phishing Websites:

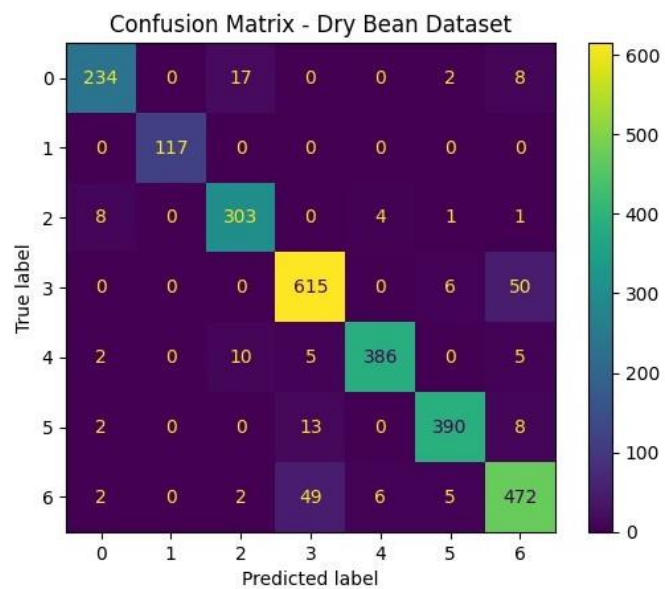
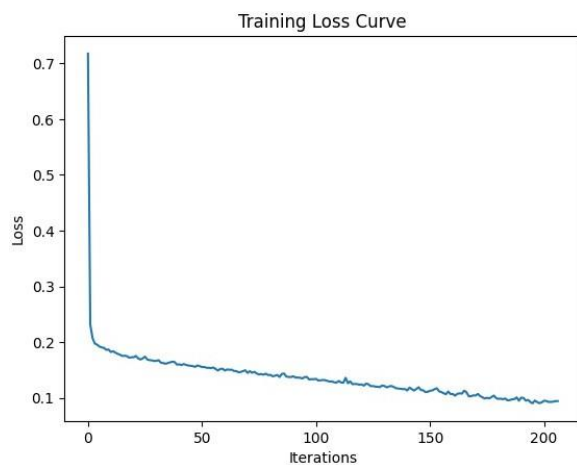
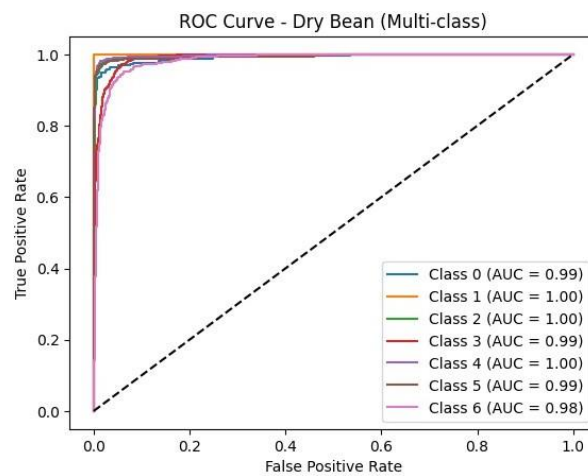
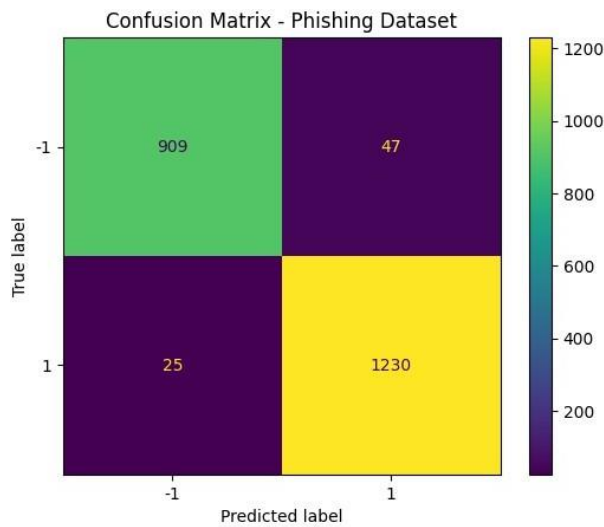
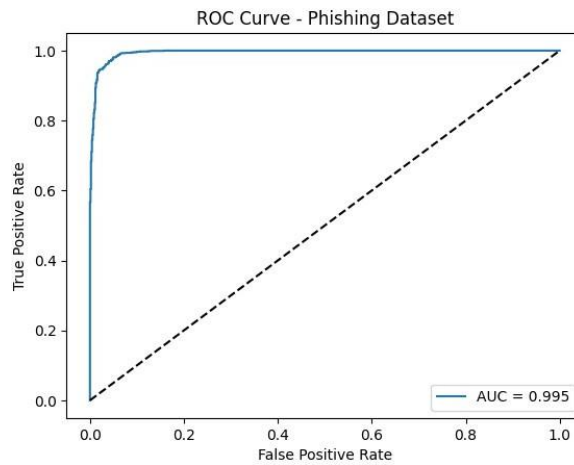
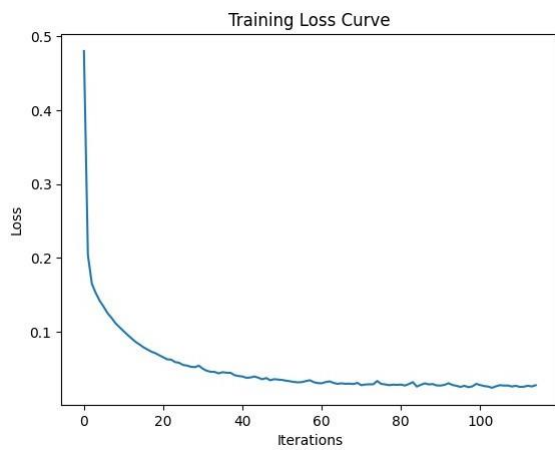
Accuracy: 0.9674355495251018

Classification Report:

	precision	recall	f1-score	support
-1	0.97	0.95	0.96	956
1	0.96	0.98	0.97	1255
accuracy			0.97	2211

ROC AUC (Dry Bean) per class: 0.98-1.00

ROC AUC (Phishing): 0.995



RESULT

The MLP Neural Network achieved 92.43% accuracy on the Dry Bean dataset and 96.74% on the Phishing Websites dataset. High AUC values (0.98–1.00) across all classes confirm excellent discriminatory power of the neural network model.